Eris Jacey Masagca

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Professional Summary

Senior backend software engineer with 3+ years building and maintaining high-scale payment systems at Visa. Proven track record in API development, database optimization, and AI integration. Currently own transaction control systems processing millions of requests daily with 99.999% availability. Champion of innovative AI-driven solutions that reduce operational overhead by 90% while maintaining production excellence.

Technical Skills

Languages: Java, JavaScript, Python, TypeScript, SQL, C, C++, Golang, Rust, Haskell
Frameworks & Technologies: Spring Boot, React, Node.js, MongoDB, Kafka, Spring AI, Model Context Protocol
Tools & Platforms: Git/GitHub, Jenkins, Docker, Prometheus/Grafana, Jira/Confluence

Work Experience

Visa Inc. Singapore
Senior Software Engineer, Value-Added Services (Transaction Services) Jul 2026 – Present
Software Engineer, Value-Added Services (Transaction Services) Jun 2023 – Jun 2026

- Own and maintain critical Visa Transaction Controls and Visa Flexible Credentials APIs integrated with VisaNet, serving issuer applications and admin portals with sub-300ms response times to support 99.999% system availability
- Led development of real-time operational analytics platform providing granular DB performance metrics for transaction APIs, reducing incident remediation time by 50%+ through pre-emptive anomaly detection during peak seasons
- Executed MongoDB migration from replica sets to sharded clusters, supporting scale from 300M to 5B+ cards while maintaining zero downtime during transition
- Pioneered AI integration across product suite: deployed production AI feature validating merchant category codes, cutting client onboarding time by 90%; developed Model Context Protocol server for internal debugging efficiency
- Developed and maintained React-based admin portal features for Visa and issuer CSRs to perform VTC/VFC operations, improving user experience and operational efficiency
- Triaged and resolved critical production incidents with VisaNet timeout errors, coordinating emergency change requests with SRE and DevOps teams to maintain system reliability
- Mentored 3 new engineers as onboarding buddy, achieving <2 week time-to-first-commit through structured knowledge transfer

Visa Inc. Singapore
Software Engineering Intern, VAS (Issuing Solutions) Payment Digitization Platform Team May 2022 – Aug 2022

- Developed RESTful API features using Spring Boot for payment digitization services, contributing to production codebase

National University of Singapore Singapore
Teaching Assistant (CS2030S, CS1010E, CS1101S) Aug 2020 – May 2022

- Facilitated weekly tutorial sessions for 8-23 students across 4 semesters; received 11 teaching award nominations with average 4.5/5.0 instructor rating

Education

National University of Singapore Singapore
Bachelor of Computing (Computer Science) with Honours (Highest Distinction) GPA: 4.61/5.00 | 2019 – 2023
Relevant Coursework: Software Engineering, Database Systems, Parallel & Concurrent Programming, Data Structures & Algorithms, Programming Language Concepts
Academic Achievement: 11 teaching award nominations across 4 semesters as TA; conducted Java workshop for 400 students

Additional Technical Experience

Key Technical Competencies

- Production Systems: Designed and maintained RESTful APIs handling millions of daily requests with strict SLA requirements (<300ms response time); experienced in incident management, root cause analysis, and performance optimization
- Database Architecture: Deep expertise in MongoDB operations including sharding strategies, replica set management, query optimization, and index design for high-scale applications
- AI/ML Integration: Hands-on experience productionizing AI features using Spring AI and MCP; implemented LLM-powered validation systems reducing manual workflows by 90%
- Cross-functional Leadership: Collaborated with product management, architects, DevOps, SRE, and security teams to deliver features aligning technical feasibility with business requirements